



Contextual Report

BSc (Hons) Software Engineering
CIS013-3 Research Methodologies and Emerging
Technologies

Kavindu Nethvitha Malshan Piladuwa Loku Bogahawatte
2541047

Date of Submission: 25th April 2026

Table of Contents

List of Figures	2
1. Introduction.....	3
1.1 Project Background	3
1.2 Aim and Objectives.....	3
1.3 Description of the Artefact / Project Scope	4
1.4 Planning on How the Problem Will Be Solved	4
2.1 Literature Review	5
2.1.1 Introduction	5
2.1.2 Kelani River Basin Overview	5
2.1.3 Water Level Monitoring in Kelani River	5
2.1.4 2016 Kelani River Flood Event	6
2.1.5 Dittwah Cyclone Flood Event (2025)	6
2.1.6 Existing Flood Warning Systems in Sri Lanka.....	7
2.1.7 Flood Prediction Using Rainfall Data	7
2.1.8 Research Gap.....	8
2.1.9 Proposed Solution (Project Idea)	8
2.1.10 Conclusion.....	8
2.1.11 References	9
2.2 Market Research.....	9
2.2.1 Introduction	9
2.2.1 Data Gathering Method.....	9
2.2.2 Requirement Analysis	10
2.2.3 Conclusion.....	15
3. Project Plan	15
3.1 Project Description.....	15
3.2 Project Tasks and Sub Tasks	17
3.3 Milestones	18
3.4 Gantt Chart (Project Timeline - 12 Weeks)	19
3.5 Summary	19
4. Planning System Artifact	19
4.1 Methodology (Agile Method).....	19
4.2 Planning.....	20
4.3 Requirement Gathering and Analysis.....	20
4.4 Design	21
4.5 Implementation.....	23
4.6 Testing and Evaluation Strategy.....	24
4.7 Risks and Risk Mitigation	24

5. References	25
6. Appendices	27
Appendix A - Work Breakdown Structure (WBS)	27
Appendix B - Survey Form (Google Form)	28
Appendix C - Survey Responses	28
Appendix D - Gantt Chart (Project Plan)	28

List of Figures

FIGURE 2.1: SURVEY RESULTS SHOWING THE AREA OF RESIDENCE OF PARTICIPANTS IN THE KELANI RIVER BASIN	10
FIGURE 2.2: SURVEY RESULTS SHOWING HOW CLOSE PEOPLE LIVE TO THE KELANI RIVER	11
FIGURE 2.3: SURVEY RESULTS SHOWING WHETHER PEOPLE HAVE EXPERIENCED FLOODING	11
FIGURE 2.4: SURVEY RESULTS SHOWING PROBLEMS FACED DURING FLOODS	12
FIGURE 2.5: SURVEY RESULTS SHOWING HOW PEOPLE RECEIVE FLOOD WARNINGS	13
FIGURE 2.6: SURVEY RESULTS SHOWING PREFERRED METHODS TO RECEIVE FLOOD ALERTS	13
FIGURE 2.7: SURVEY RESULTS SHOWING TIME NEEDED TO PREPARE AFTER RECEIVING A WARNING	14
FIGURE 2.8: SURVEY RESULTS SHOWING WHAT INFORMATION PEOPLE NEED IN FLOOD ALERTS	14
FIGURE 3.1: GANTT CHART SHOWING PROJECT SCHEDULE AND TASK TIMELINE	19
FIGURE 6.1: WBS STRUCTURE SHOWING MAIN ACTIVITIES AND SUB ACTIVITIES OF THE PROJECT	27

List of Tables

TABLE 3.1: PROJECT MILESTONES OF THE FLOOD DETECTION AND ALERT SYSTEM	18
---	----

1. Introduction

1.1 Project Background

Flooding is a common and serious problem in Sri Lanka, especially in areas near the Kelani River. The Kelani River flows through places like Colombo where many people live and work. Because of this, floods can damage houses, roads, and shops. When there is heavy rain, the river water level can go up quickly and overflow, causing floods.

Flooding has happened many times in Sri Lanka. Some areas near the Kelani River get floods again and again, especially during the rainy season. One big example is the 2016 flood. It caused a lot of damage in many areas. Many people lost their homes, roads were blocked, and daily life was badly affected.

Another recent example is the Ditwah cyclone. It brought heavy rain and caused floods in Colombo and nearby areas. Many houses were damaged, transport was disturbed, and people faced many problems. These examples show that flooding is not a one time problem. It happens again and again and affects many people.

Right now, flood warnings in Sri Lanka are given through TV, radio, and public announcements. These are helpful, but they are not always fast. Some people in danger areas do not get the warning on time. Because of this, they may not be able to protect themselves or their property.

Today, new technology can help solve this problem. Sensors can be used to check water levels, and weather data can be used to see how much it is raining. This information can help to find flood risks early. Also, websites and mobile apps can send warning messages directly to people who live in risky areas.

This project is about creating an Automatic Early Flood Detection and Warning System for the Kelani River. The system will collect real time water level and rainfall data, check the data, and send early warning messages to users based on their location. The system will have a website and an Android mobile app, so it is easy to use.

The main goal of this project is to help people stay safe and reduce damage from floods. By sending early warnings directly to users, people can take action quickly. This system can also help improve communication between authorities and the public. So, this project is important for protecting people who live in flood risk areas of Sri Lanka.

1.2 Aim and Objectives

Aim

The aim of this project is to develop an Automatic Early Flood Detection and Warning System for the Kelani River that can collect real time data, detect possible flood risks, and send early warning messages to people in affected areas.

Objectives

- To collect real time water level data from the Kelani River using sensors.
- To collect rainfall and weather data from online sources such as weather APIs and government websites.
- To analyse the collected data to identify possible flood risks early.
- To design and develop a user friendly website to monitor flood data and manage users.
- To develop an Android mobile application using WebView for easy access to the system.
- To create a system that can send automatic warning messages (emails) to users based on their location.
- To store user and system data safely using a database.
- To test the system using real time and historical data to check its accuracy and performance.
- To improve communication between authorities and the public by providing fast and direct alerts.

1.3 Description of the Artefact / Project Scope

The artefact of this project is an Automatic Early Flood Detection and Warning System for the Kelani River. This system will include a website and an Android mobile application (WebView app). It will also use a database, sensors, and an automatic alert system.

The main purpose of this system is to check real time data like river water level and rainfall. Then it will analyse the data to find possible flood risks early. If a flood risk is found, the system will send warning messages to people who are registered, based on their location.

The system will include these parts,

- A website to show flood data and manage users
- An Android mobile app for easy access
- A database to store user and system data
- Water level sensors to collect river data
- Weather API to get rainfall and weather information
- An email system to send warning messages

This project will focus only on the Kelani River area in Sri Lanka. It will only detect floods and send warnings. It will not control or stop floods.

1.4 Planning on How the Problem Will Be Solved

The problem of late flood warnings and lack of real time information will be solved using sensors, data analysis, and automatic alerts. The system will work step by step as follows:

Collecting Data

- Water level sensors will collect river water data. Weather data like rainfall will be taken from online sources and government websites.

Checking and Analysing Data

- The system will check the collected data. If water level or rainfall is too high, the system will understand that there is a flood risk.

Building the System

- A website will be created to show live data and manage users. An Android app will be made using WebView so users can easily use the system on mobile phones.

Storing Data

- A database will store user details and location information safely. Users will be grouped based on their area for sending alerts.

Sending Alerts

- If a flood risk is found, the system will automatically send email warnings to users in the affected areas. This will help people get warnings quickly.

Testing the System

- The system will be tested using old flood data and simulated situations. This is to check if the system works correctly and gives accurate warnings.

2.1 Literature Review

2.1.1 Introduction

Flooding is a big problem in Sri Lanka, especially in areas near the Kelani River. Many people lose houses, roads, and daily income when floods happen. Because of this, many researchers and government groups study floods to understand why they happen and how to reduce damage.

This literature review looks at past studies, reports, and real flood events. It focuses on the Kelani River, water levels, rainfall, and warning systems. The main aim is to understand how floods happen and what problems are in current systems.

This review uses information from past years, especially the 2016 flood and the recent 2025 Ditwah cyclone. The sources include research papers and government websites like the Department of Meteorology, Irrigation Department, and Disaster Management Centre.

2.1.2 Kelani River Basin Overview

The Kelani River is one of the largest rivers in Sri Lanka. It starts from the central hill country and flows through many towns before reaching the sea near Colombo.

The river basin is very important because many people live in this area. It also supports agriculture, transport, and water supply.

The Kelani River has several main sub rivers,

- Kehelgamu Oya
- Maskeli Oya
- Seethawaka Ganga

Kehelgamu Oya and Maskeli Oya join together near Kithulgala and form the main Kelani River.

Maskeli Oya is connected with important reservoirs such as,

- Laxapana Reservoir
- Canyon Reservoir
- Maussakele Reservoir

These reservoirs are used for hydro power and water storage. They also affect river water flow during heavy rain.

Seethawaka Ganga joins the Kelani River in the middle area and increases water flow during rainy periods.

Because of these rivers and reservoirs, water movement in the Kelani River is very complex. Flooding can happen in different ways depending on rainfall and water release.

2.1.3 Water Level Monitoring in Kelani River

The Sri Lankan government monitors the Kelani River using water level stations. These stations help to measure river height and rainfall.

Important monitoring stations include,

- Norton
- Kithulgala
- Deraniyagala
- Holombuwa
- Glencourse
- Hanwella
- Nagalagam Street

These stations are placed from the upper river to the lower river,

For example,

- Norton and Kithulgala monitor upper river conditions
- Deraniyagala and Holombuwa monitor middle basin rainfall and water flow
- Hanwella and Nagalagam Street monitor flood conditions in lower areas

These stations send data to government departments like the Irrigation Department.

However, most systems only collect and display data. They do not send direct automatic alerts to people in real time.

2.1.4 2016 Kelani River Flood Event

The flood in May 2016 was one of the worst floods in Sri Lanka after 1989.

Heavy rain continued for many days across the Kelani River basin. The total rainfall in some areas was very high. Water levels increased quickly in rivers and streams.

According to studies, the rain was not only heavy but also spread across many areas of the basin. This caused water to rise in many places at the same time.

Important findings from this event,

- Flood water rose very fast
- Some areas were flooded within a few hours
- Flood stayed for a long time in low areas
- Damage to houses and roads was very high

Even though the water level was not the highest recorded in history, the long flooding time caused more damage. This event showed that early warning is very important, because people had very little time to react.

2.1.5 Ditwah Cyclone Flood Event (2025)

A more recent event happened in November 2025 during the Ditwah cyclone.

This event showed a different flood pattern compared to 2016.

In this case,

- Heavy rain happened mainly in upper basin areas
- Lower basin areas did not get much rain
- Still, flooding happened in lower areas

This happened because reservoirs like Laxapana and Canyon filled with water. When they reached full capacity, excess water had to be released.

The sudden release of water moved quickly to lower river areas. This caused flooding even without local rainfall.

This shows an important fact:

Flooding in the Kelani River is not only caused by local rain. It also depends on upstream rainfall and reservoir water release.

2.1.6 Existing Flood Warning Systems in Sri Lanka

In Sri Lanka, flood warnings are mainly given by government departments. They use,

- Television announcements
- Radio messages
- Public warnings
- Government notices

These methods help to inform people, but they also have problems,

- Messages are slow
- Not all people receive warnings
- No personal warning system
- No mobile app alerts
- No location based warning system

Because of these problems, many people still receive warnings late.

Some modern countries use IoT sensors, mobile apps, and automatic alert systems. These systems give fast warnings before floods happen.

Sri Lanka still has limited use of these modern systems in the Kelani River basin.

2.1.7 Flood Prediction Using Rainfall Data

Research studies show that rainfall is strongly connected to river water levels.

When heavy rain happens in upper areas of the river, water flows down and increases river levels in lower areas.

There is also a time delay between rainfall and flood occurrence.

Because of this, future rainfall data can be used to predict flood risk.

In modern systems, rainfall data can come from,

- Weather APIs
- Satellite data
- Government weather stations

By using this data, it is possible to estimate how much river water level will increase in the future.

This helps to detect flood risk before it happens.

2.1.8 Research Gap

From all studies and reports, some important problems are found,

- No full real time prediction system for Kelani River
- No system using future rainfall to predict water level
- No system combining rainfall, river level, and reservoir data
- No fast automatic alert system for public users
- No system giving location based flood warnings

Most current systems only show data, but do not predict floods early enough.

Because of this, people still receive warnings late and face damage.

2.1.9 Proposed Solution (Project Idea)

To solve these problems, this project proposes an Early Flood Detection and Warning System for the Kelani River.

The system will,

- collect real time river water level data
- collect rainfall data from weather APIs
- use future rainfall to predict water levels
- check upper river and reservoir conditions
- detect flood risk early
- send automatic email alerts
- show data on a website
- provide Android WebView application

The system will send warnings before floods happen.

It will also send alerts only to users in affected areas.

This will help people prepare early and reduce damage.

2.1.10 Conclusion

Flooding in the Kelani River is a serious and repeated problem in Sri Lanka.

Past studies and real events show that floods depend on rainfall, river structure, and reservoir water release.

The 2016 flood showed fast and wide flooding with short warning time. The 2025 Ditwah cyclone showed that reservoir release can also cause flooding even without local rain.

Current warning systems are not fast or advanced enough. They do not use prediction models or real time alerts properly.

Therefore, a better system is needed.

This project will improve flood safety by using rainfall prediction, water level prediction, and automatic alert systems.

2.1.11 References

- Hettiarachchi, P. (2016) *Hydrological Report on the Kelani River Flood in May 2016*. Colombo: Irrigation Department, Sri Lanka.
- Hettiarachchi, P. (2018) *Expansion and Modernization of Hydro Meteorological Information System (HMIS) of Sri Lanka, Under the Dam Safety and Water Resources Planning Project (DSWRPP)*. Available at: ResearchGate (Accessed: 23 April 2026).
- Hettiarachchi, P. (2019) *Flood Frequency Analysis and Selection of Suitable Design Flood for Kelani Flood Bunds*. Available at: ResearchGate (Accessed: 23 April 2026).

2.2 Market Research

2.2.1 Introduction

Market research was done to understand the real situation of people living near the Kelani River and how floods affect their daily life. It also helps to understand how people currently receive flood warnings and what kind of warning system they need.

The Kelani River basin is one of the most flood affected areas in Sri Lanka. Every year, many families face floods due to heavy rain and rising river water levels. These floods cause serious problems such as house damage, transport issues, health problems, no electricity, and loss of income.

Because of these problems, it is important to study real user needs before designing the system. This research helps to find out the weaknesses in the current warning methods and what improvements are needed.

2.2.1 Data Gathering Method

The data was collected using a Google Form survey. The survey included 10 simple questions written in both Sinhala and English. This was done so that all people could easily understand and answer without any language problems. The survey was shared with people living in different areas of the Kelani River basin. A total of 43 responses were collected.

Population

The population of this study is:

- All residents living in the Kelani River basin area
- Flood prone areas such as Hanwella, Kelaniya, Biyagama, Malwana, Homagama, Padukka, and nearby regions

Sample

The sample of this study is:

- 43 residents from different flood affected locations in the Kelani River basin

These people were selected because:

- They live near or around the Kelani River
- They have experienced floods or flood related problems
- They understand real life flood situations

This sample helps to understand real needs, problems, and expectations of people living in flood risk areas.

2.2.2 Requirement Analysis

This section explains what the survey found in very simple way.

1. Area of Residence

This question was asked to know where the respondents live. It helps to check if the survey is taken from real flood affected areas.

The results show that most people are from,

- Hanwalla
- Kelaniya
- Biyagama
- Padukka
- Malwana
- Homagama and nearby areas

This shows that the survey is correctly collected from Kelani River flood prone communities, so the data is valid.

Area of Residence / ඔබ ජීවත් වන ප්‍රදේශය
42 responses

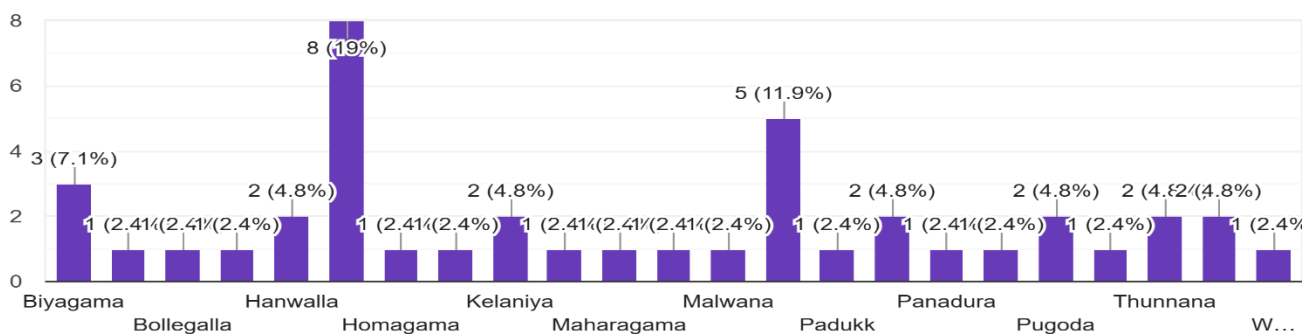


Figure 2.1: Survey results showing the area of residence of participants in the Kelani River basin

2. Distance from Kelani River

This question was asked to understand how close people live to the river and their flood risk level.

The results show,

- 16.3% live very close (within 1 km)
- 55.8% live at a medium distance

- 27.9% live far from the river

This shows that many people are still living near flood risk areas. Even people who are not very close can still be affected during heavy rains and floods.

How close are you to Kelani River? / ඔබ කැලණි ගඟට කොපමණ දුරද?
43 responses



Figure 2.2: Survey results showing how close people live to the Kelani River

3. Flood Experience

This question was asked to check whether people have experienced floods before.

The results show,

- 19 people were directly affected by floods
- 20 people were indirectly affected (road blocking, transport issues, etc.)
- Only 4 people were not affected

This shows that almost everyone in the area has faced flood related problems in some way.

Have you experienced flooding or been affected by flood related situations (e.g., road blocks, transport issues)? ඔබගේ ප්‍රදේශයේ ගංවතුර හේතුවෙන්...දැමීම, ප්‍රවාහන ගැටලු) බලපෑම් අත්විඳි තිබේද?
43 responses

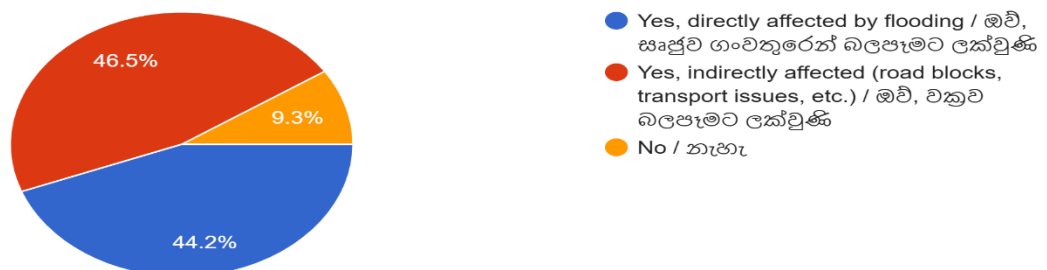


Figure 2.3: Survey results showing whether people have experienced flooding

4. Problems During Floods

This question was asked to find the main problems people face during floods.

The main problems are,

- Transport problems (roads blocked, cannot travel)
- No electricity and water supply
- Damage to houses and property
- Health problems (diseases, injuries)
- Loss of income (cannot go to work)

This shows that floods affect almost every part of daily life.

What problems did you face during floods? / ගංවතුර කාලයේ ඔබට මුහුණ දීමට සිදු වූ ගැටලු මොනවාද?
43 responses

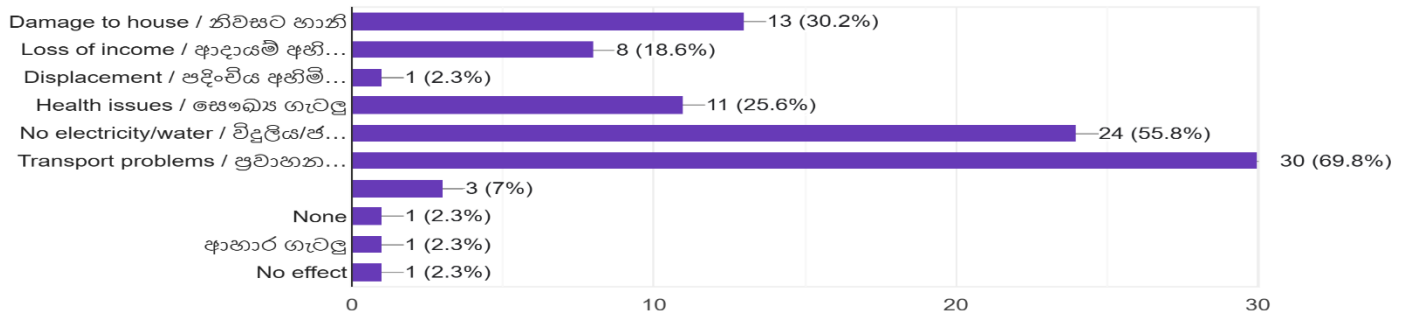


Figure 2.4: Survey results showing problems faced during floods

5. Flood Warning Methods

This question was asked to understand how people currently receive flood warnings.

The results show,

- Social media is the most used method
- TV and SMS are also used
- Radio is used by some people
- Some people do not receive any warning

This shows that current warning systems are not fully reliable and not everyone receives alerts on time.

Do you receive flood warnings? If yes, how do you receive them? ඔබට ගංවතුර අනතුරු ඇඟවීම් ලැබේද? ලැබේ නම්, ඒවා ලැබෙන්නේ කෙසේද?

43 responses

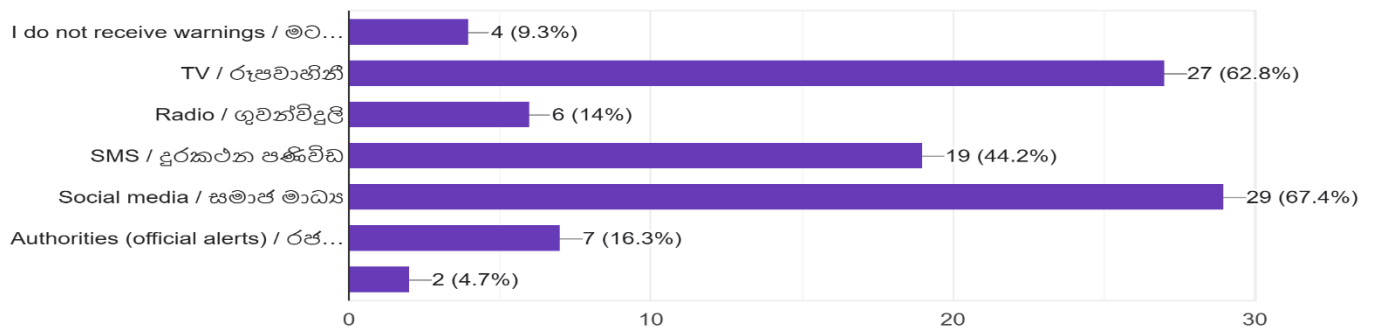


Figure 2.5: Survey results showing how people receive flood warnings

6. Preferred Warning Method

This question was asked to understand how people want to receive flood alerts in the future.

The results show,

- SMS alerts are the most preferred
- Mobile app notifications are also highly preferred
- TV alerts are used by some people
- Siren or loudspeaker alerts are also useful

This shows that people prefer fast and direct warning methods, especially mobile based alerts.

How would you prefer to receive flood alerts? ඔබට ගංවතුර අනතුරු ඇඟවීම් ලැබීමට කැමති ආකාරය කුමක්ද?

43 responses

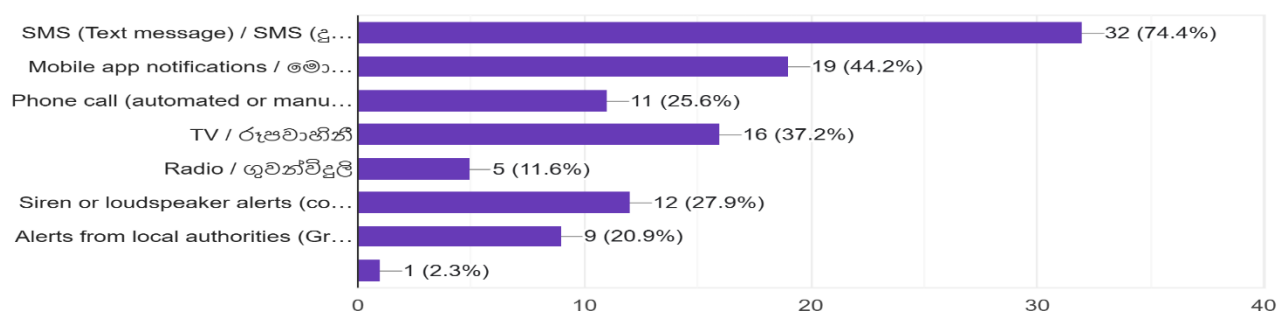


Figure 2.6: Survey results showing preferred methods to receive flood alerts

7. Time Needed After Warning

This question was asked to understand how much time people need after receiving a flood warning.

The results show:

- 1 - 3 hours - most people
- More than 3 hours - some people
- Less than 30 minutes - few people

This shows that people need early warnings, so they have enough time to move to safety, protect their belongings, and prepare properly.

After receiving a flood warning, how much time do you need to prepare and take necessary actions? ගංවතුර අනතුරු ඇඟවීමක් ලැබූ පසු ඊයාමාර්ග ගැනීමට ඔබට අවශ්‍ය කාලය කොපමණද?
43 responses

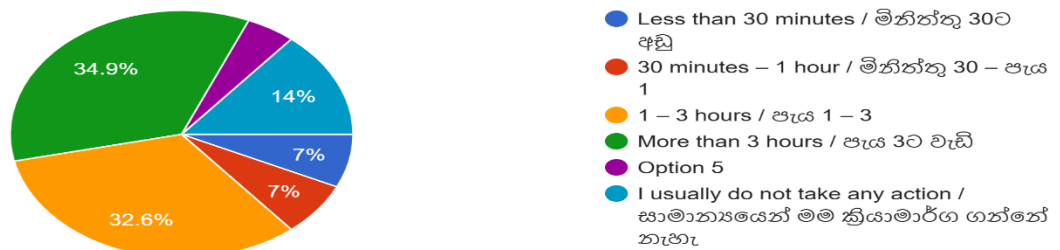


Figure 2.7: Survey results showing time needed to prepare after receiving a warning

8. Required Information in Flood Alerts

This question was asked to find out what type of information people need in a flood warning message.

The survey results show the following responses (43 total),

- Forecasted water level - 20 responses
- Expected flood timing - 16 responses
- Flood risk level (low / medium / high) - 33 responses

What information do you need in a flood alert? (Select all that apply) ගංවතුර අනතුරු ඇඟවීමක නිව්ය යුතු තොරතුරු මොනවාද? (අදාළ සියල්ල තෝරන්න)
43 responses

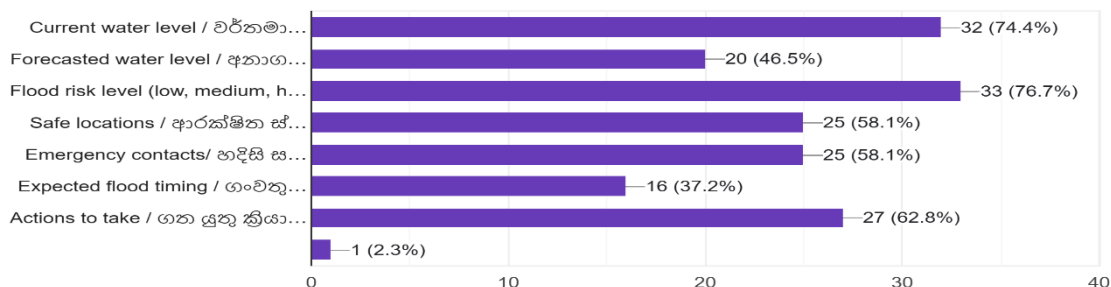


Figure 2.8: Survey results showing what information people need in flood alerts

These results show that people mostly want future prediction information, not only current data.

The most important needs are,

- Forecasted water level

- Expected flood timing
- Flood risk level

This means people want to know what will happen in the future, not only what is happening now.

2.2.3 Conclusion

The survey clearly shows that flooding is a frequent and serious problem in the Kelani River basin. Most people (over 90%) have been affected by floods directly or indirectly. Many areas such as Hanwella, Kelaniya, Biyagama, Padukka, and Malwana are repeatedly impacted.

The current warning system is not fully effective, because,

- Some people do not receive warnings
- Information is delayed or unclear
- Many people rely only on social media or TV

The results also show that people strongly prefer,

- SMS alerts
- Mobile app notifications
- Fast and clear warning messages with location based information

Research Gap Identified

From the survey and analysis, the main gaps in existing systems are,

- No proper real time water level prediction system
- No location based automatic warning system
- No system that combines rainfall + river level + prediction model
- Delayed communication of flood risk to local people

How This Project Solves the Problem

The proposed system will,

- Collect real time water level and rainfall data
- Use a prediction model to forecast flood risk early
- Send automatic alerts via SMS/email/mobile app
- Give clear information (risk level, timing, safety actions)
- Focus on specific affected locations in Kelani River basin

This will help people get earlier, faster, and more accurate flood warnings, which can reduce damage and save lives.

3. Project Plan

3.1 Project Description

This project is called “Implementing an Early Flood Detection and Warning System for the Kelani River”. The main goal of this system is to reduce flood damage by giving early warning messages before flooding happens.

The system will work by collecting and studying,

- Past rainfall data in the Kelani River basin
- Past river water level data
- Historical flood records (example: 1989 flood, 2016 flood, recent events)
- Flood affected areas in the Kelani River basin

This information will help to understand how rainfall affects river water levels and how floods start and spread in different locations.

Data Collection and Dataset Creation

The first important step of this project is building a dataset. This dataset will contain,

- Rainfall data (past and real time)
- River water level data
- Flood history data
- Location based risk areas (high, medium, low risk zones)

This dataset will help the system to study patterns between rainfall and water level changes.

System Function

After building the dataset, the system will,

- Analyze rainfall and water level changes
- Detect rising flood risk
- Predict possible flood situations
- Send early warning alerts to users

The system will send alerts using,

- Email
- Mobile notifications (Android app)
- Web dashboard messages

Flood Risk Identification

The system will also identify flood prone areas such as,

- Hanwella
- Kelaniya
- Biyagama
- Malwana
- Kaduwela
- Other low lying areas in the Kelani River basin

People in these areas will receive alerts first.

System Development Approach

The system will be developed step by step using Agile method,

- Each part will be developed and tested separately
- Feedback will be used to improve the system
- Final system will combine all parts together

3.2 Project Tasks and Sub Tasks

Task 1 Requirement Analysis

- Study Kelani River flood problems
- Collect past flood and rainfall information
- Identify system needs
- Analyze survey results

Task 2 Dataset Preparation

- Collect historical rainfall data
- Collect river water level data
- Collect flood history data
- Combine all data into one dataset
- Clean and organize data for analysis

Task 3 System Design

- Design system architecture
- Design database structure (MySQL)
- Create flowcharts
- Design web and mobile UI layouts

Task 4 Backend Development

- Create PHP backend system
- Develop APIs for data handling
- Create user registration system
- Connect database with system

Task 5 Frontend Development (Web System)

- Develop admin dashboard
- Display water level and rainfall data
- Show flood risk information
- Create user interface using HTML, CSS, JavaScript

Task 6 Android Application Development

- Create Android WebView application
- Connect mobile app to web system
- Show alerts and updates on mobile

Task 7 Data Integration

- Connect weather API (real time rainfall data)

- Connect river level data sources
- Store live data in database

Task 8 Flood Prediction System

- Analyze rainfall and water level data
- Detect flood risk levels
- Create simple prediction model
- Identify early flood signals

Task 9 Alert System

- Send email alerts to users
- Send mobile notifications
- Show alerts on dashboard
- Location based warning system

Task 10 Testing

- Test each module separately
- Check system accuracy
- Fix errors and bugs
- Final system testing

Task 11: Documentation

- Write final report
- Add diagrams and screenshots
- Explain system functions
- Prepare final submission

3.3 Milestones

Milestone	Description
M1	Project topic approved
M2	Requirement analysis completed
M3	Dataset collected and prepared
M4	System design completed
M5	Backend and database completed
M6	Web system completed
M7	Android app completed
M8	Prediction and alert system working
M9	Testing completed
M10	Final report submitted

Table 3.1: Project milestones of the Flood Detection and Alert System

3.4 Gantt Chart (Project Timeline - 12 Weeks)



Figure 3.1: Gantt chart showing project schedule and task timeline

3.5 Summary

This project is planned from May 01, 2026 to August 01, 2026 following a clear and structured approach.

At the beginning, rainfall data, water level data, and past flood information related to the Kelani River are collected and analyzed. Based on this data, a dataset is prepared to support the development of a flood prediction model.

After that, the system is developed step by step, including backend development, web interface, and mobile integration. The system is designed to predict flood risk and provide early warnings to users through methods such as SMS and mobile notifications.

Finally, the system is tested to ensure accuracy and proper functionality. The project is then completed with proper documentation and submitted within the given deadline.

This structured plan ensures that the system is well-organized, reliable, and suitable for real world flood detection and early warning.

4. Planning System Artifact

4.1 Methodology (Agile Method)

This project will use the Agile method. System will be built step by step in small parts called sprints.

In each step,

- The work will be planned
- The system will be designed
- The features will be developed
- The system will be tested

After each step,

- Errors will be fixed
- Improvements will be added

This method will help to,

- Find problems early
- Improve the system step by step

4.2 Planning

The project plan was created for the period May 1 to August 1 and this work was divided into main stages,

- Requirement analysis
- Data collection
- System design
- Development
- Testing
- Final report

Each task was given,

- Start date
- End date
- Priority level
- Due date

A Gantt chart was created to,

- Track progress
- Manage time
- Complete tasks on time

This planning helped to organize the project clearly.

4.3 Requirement Gathering and Analysis

Requirements will be gathered using,

1. Survey

A Google Form survey was already conducted.

- 43 responses were collected
- Questions were in Sinhala and English
- People were from Kelani River areas

This helped to understand real user problems.

2. Literature Review

Past studies and reports will be used to understand,

- Flood causes

- Rainfall patterns
- Water level changes
- Past flood events

3. Real Flood Situations

Real flood events will be analyzed, such as,

- 2016 Kelani flood
- Recent flood situations
- Reservoir water release issues

User Requirements

Based on analysis, users need,

- Early flood alerts
- Future water level prediction
- Flood risk levels
- Safe locations
- Actions to take

System Requirements

The system will,

- Collect rainfall data
- Collect water level data
- Predict future water levels
- Detect flood risk
- Send alerts to user

4.4 Design

The system will be designed as follows

4.4.1. Dataset Design

A dataset will be created using,

- Past rainfall data
- Past water level data
- Flood history data
- Reservoir data

This dataset will be used to,

- Train the prediction model
- Analyze patterns

- Improve accuracy

4.4.2. System Design

Input,

- Rainfall data
- Water level data

Process,

- Data will be analyzed
- Water levels will be predicted
- Flood risk will be calculated

Output,

- Alerts
- Warning messages

4.4.3. Device and Data Sources

In this project, data will be collected in different ways.

4.4.3.1. Government Water Level Stations (Future Option)

There are official water level stations in the Kelani River, such as,

- Norton
- Kithulgala
- Holombuwa
- Glencourse
- Hanwella
- Nagalagam Street
- Deraniyagala

These stations measure real water levels.

However,

- This project cannot directly access these stations now
- Special permission is needed

Future Plan,

If permission is given, the system will connect to these stations and get real time data.

4.4.3.2. Ultrasonic Sensor (Main Method in This Project)

Since real station data is not available, this project will use an ultrasonic sensor to measure water level.

4.4.3.3. Rainfall Data Sources

Rainfall data will be collected from,

- Online weather websites
- Weather APIs

This data is important because,

- Rainfall affects water level
- Helps to predict floods

4.4.3.4. Mobile Devices

Mobile phones will be used to,

- Receive flood alerts
- Show warning messages

The system will send alerts using,

- SMS /Mails
- Mobile app notifications

4.5 Implementation

The system will be developed using:

Technologies

- Backend - PHP
- Database - MySQL
- Web frontend - HTML, CSS, JavaScript and their Frameworks
- Mobile App - Android, Java, Kotlin

System Features

The system will include,

- Data collection module
- Data processing module
- Prediction system

- Alert system

4.6 Testing and Evaluation Strategy

4.7 Risks and Risk Mitigation

This section explains possible problems in the system and how they will be reduced or solved.

Risk 1 - Incorrect Data

- Problem - Wrong or incomplete data can lead to wrong flood predictions.
- Solution - Use trusted data sources such as government stations and verified weather data.

Risk 2 - System Errors

- Problem - The system may crash or stop working.
- Solution - Do proper testing (white box and black box testing) and fix errors early.

Risk 3 - Late Alerts

- Problem - Flood warnings may come too late.
- Solution - Use a fast alert system with SMS and mobile notifications.

Risk 4 - Network Issues

- Problem - Users may not receive alerts due to poor internet or mobile signal.
- Solution - Use multiple channels such as SMS, mobile app, and notifications.

Risk 5 - Low Accuracy

- Problem - Flood prediction may not be fully correct.
- Solution - Improve the model by using more rainfall and water level data over time.

Risk 6 - Server Down (System Failure)

- Problem - If the server goes down, the system will stop working and alerts will not be sent.
- Solution - Use backup server options, cloud hosting, and automatic system recovery methods.

Risk 7 - No Real time Data Availability

- Problem - Sometimes real-time water level data from stations may not be available or delayed.
- Solution - Use alternative data sources like rainfall data, historical data, and sensor based estimation until data is restored.

5. References

- Arumugam, S. (1969) *Water Resources of Ceylon: Its Utilization and Development*. Colombo: Water Resources Board.
- Asian Development Bank (2019) *Sri Lanka Disaster Risk Profile*. Manila: ADB.
- Bates, P.D. et al. (2008) 'Flood modelling', *Journal of Hydrology*.
- Beven, K. (2012) *Rainfall-Runoff Modelling: The Primer*. Chichester: Wiley.
- Central Environmental Authority Sri Lanka (2017) *Flood Impact Assessment Report*. Colombo: CEA.
- Chen, J. et al. (2017) 'Real-time flood monitoring systems', *Sensors*.
- Chow, V.T., Maidment, D.R. and Mays, L.W. (1988) *Applied Hydrology*. New York: McGraw-Hill.
- Colombo Municipal Council (2017) *Urban Flood Management Plan*. Colombo: CMC.
- Darmasena, G.T. (1989) *Floods in Kelani Ganga*. Hydrological Annual 1988/89. Colombo: Irrigation Department.
- Department of Meteorology Sri Lanka (2022) *Annual Climate Report*. Colombo: DoM.
- Dissanayake, P. (2020) 'Kelani River modelling', *Sri Lanka Engineering Journal*.
- Fernando, T. (2016) 'Kelani River flood impacts', *Sri Lanka Engineering Review*.
- Gunawardena, N. (2018) 'Rainfall trends in Sri Lanka', *Meteorological Science Journal*.
- Herath, S. (2015) 'Hydrology of Sri Lanka rivers', *Journal of Hydrology Sri Lanka*.
- Hettiarachchi, P. (2016) *Hydrological Report on Kelani River Flood in May 2016*. Colombo: Irrigation Department.
- Hettiarachchi, P. (2018) *Expansion and Modernization of Hydro-Meteorological Information System (HMIS) of Sri Lanka*. DSWRPP Project.
- Hettiarachchi, P. (2019) 'Flood Frequency Analysis and Selection of Suitable Design Flood for Kelani Flood Bunds'. Available at: ResearchGate.
- IPCC (2021) *Climate Change 2021: The Physical Science Basis*. Geneva: IPCC.
- Irrigation Department Sri Lanka (2016) *Scheme of Organization and Standing Orders to Safeguard Colombo from Kelani River Floods*. Colombo.
- Irrigation Department Sri Lanka (2021/22) *Hydrological Annual of Sri Lanka*. Colombo.
- Irrigation Department Sri Lanka (2022) *Hydrometric Data Report – Kelani Basin*. Colombo.

- Jain, S.K. et al. (2018) 'Hydrological modelling methods', *Hydrological Sciences Journal*.
- Jayasinghe, A. (2020) 'Flood warning systems in Sri Lanka', *Disaster Management Journal*.
- Kundzewicz, Z.W. et al. (2014) 'Flood risk and climate change', *Hydrological Sciences Journal*.
- Liu, X. et al. (2018) 'Ultrasonic water level sensors', *Measurement*.
- Ministry of Irrigation Sri Lanka (2021) *Kelani River Basin Flood Management Plan*. Colombo.
- Mosavi, A. et al. (2018) 'Machine learning flood prediction', *Water*.
- NBRO Sri Lanka (2016) *Flood Hazard Mapping Report*. Colombo: NBRO.
- National Disaster Management Centre Sri Lanka (2020) *Flood Early Warning System Report*. Colombo: NDMC.
- Perera, S. (2017) 'Flood hazard in Colombo', *Urban Water Journal Sri Lanka*.
- Rajapaksha, R. (2020) 'GIS flood mapping in Sri Lanka', *Geoinformatics Journal*.
- Rathnayake, U. (2017) 'Flood disasters in Sri Lanka', *Natural Hazards Asia*.
- Silva, R. and Fernando, T. (2019) 'Flood mapping in Colombo', *Journal of Hydrology Asia*.
- Singh, V.P. (2014) *Hydrologic Systems*. New Delhi: Springer.
- SLLRDC (2018) *Colombo Flood Control Project Report*. Colombo: Sri Lanka Land Reclamation and Development Corporation.
- Srivastava, P. et al. (2013) 'GIS flood forecasting', *Environmental Modelling*.
- Todini, E. (2007) 'Hydrological forecasting', *Hydrology and Earth System Sciences*.
- United Nations (2020) *Global Disaster Risk Reduction Report*. New York: UN.
- UNDP Sri Lanka (2017) *Flood Resilience Programme Report*. Colombo: UNDP.
- Wijesekera, N. (2017) 'Flood behaviour in Sri Lanka', *Engineering Journal Sri Lanka*.
- World Bank (2018) *Sri Lanka Flood Risk Management Project Report*. Washington, DC: World Bank.
- World Health Organization (2017) *Flood Health Risk Guidelines*. Geneva: WHO.
- World Meteorological Organization (2011) *Manual on Flood Forecasting and Warning*. Geneva: WMO.
- World Meteorological Organization (2019) *Flood Warning Systems Guide*. Geneva: WMO.
- Zhang, Y. et al. (2020) 'IoT-based flood warning systems', *IEEE Access*.

6. Appendices

This section includes extra information used in the project. These help to support the work done in the system.

Appendix A - Work Breakdown Structure (WBS)

The project is divided into small parts to manage the work easily.

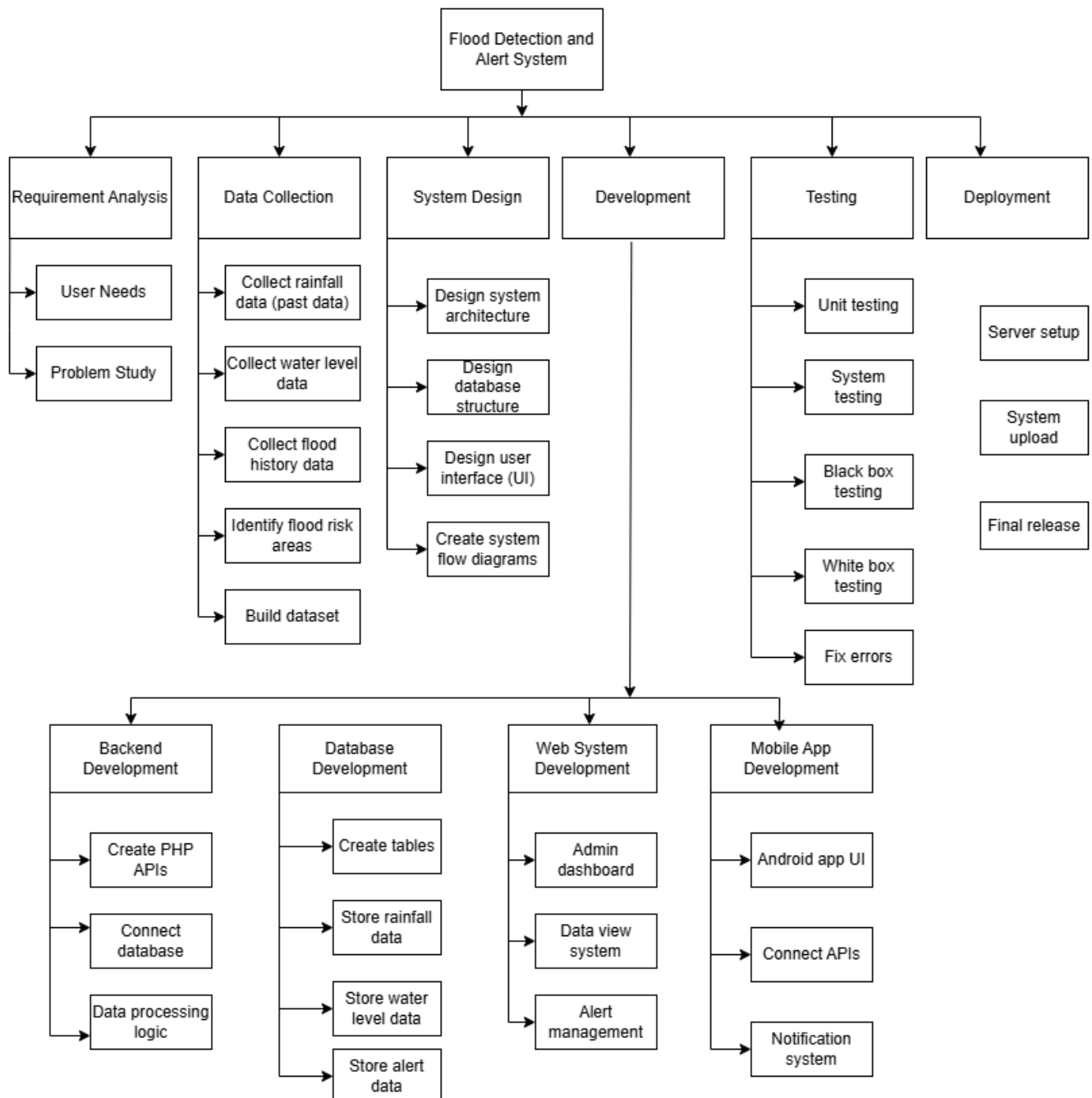


Figure 6.1: WBS structure showing main activities and sub activities of the project

Appendix B - Survey Form (Google Form)

A Google Form was created to collect user data from people living in the Kelani River basin.

Details,

- Total Questions: 10
- Languages used: Sinhala and English
- Purpose: To understand flood problems, user needs, and alert preferences

Survey Form Link,

<https://forms.gle/HdBmxudZZQJZ26Yp7>

Appendix C - Survey Responses

The responses collected from users are stored in Google Forms.

Details,

- Total Responses: 43
- Participants: Residents from flood prone areas in Kelani River basin

Survey Responses Link,

<https://docs.google.com/spreadsheets/d/1vZeIE9hYw8AOBg4fwkngccIKvIBeUSeLUoB9SEyRuuU/edit?usp=sharing>

Appendix D - Gantt Chart (Project Plan)

The project schedule was created using Monday.com.

It includes,

- Tasks and sub tasks
- Start and end dates
- Deadlines
- Priority levels
- Dependencies

Gantt Chart Project Link,

<https://infovixelits-team.monday.com/boards/5028025539>